

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is:
The web server was overwhelmed by a flood of incomplete TCP connection requests.

The logs show that:

- The attacker at IP **203.0.113.0** repeatedly sent **SYN packets** to port **443** on the web server.
- These requests never completed the three-way handshake.
- The server responded with **RST, ACK** packets and eventually timed out legitimate connections.
- Normal traffic from internal IPs (198.51.100.x) initially succeeded but later failed with **504 Gateway Time-out** errors.

This event could be:

A **TCP SYN Flood attack**, which is a type of **Denial of Service (DoS)** attack.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. **SYN**: The client sends a synchronization packet to start the connection.
2. **SYN-ACK**: The server acknowledges and responds, reserving resources for the connection.
3. **ACK**: The client confirms, completing the handshake and allowing data transfer.

Explain what happens when a malicious actor sends a large number of SYN packets all at once:

The attacker floods the server with thousands of SYN packets but never sends the final ACK. Each half-open connection consumes memory and CPU resources. The server's connection queue fills up, preventing legitimate users from completing their own handshakes.

Explain what the logs indicate and how that affects the server:

The logs show continuous SYN packets from **203.0.113.0** with no corresponding ACKs. The server eventually sends **RST, ACK** responses and legitimate users receive **504 Gateway Time-out** errors. This means the server is **overloaded**, unable to process real requests, and the website becomes **unavailable**.